Review R1: Combined Academic Review + Audit Reconciliation

"Is Volatility Targeting Just Trend Following? Decomposing the Benefits of Volatility Targeting"

Lai (2026) — body_v2.tex, ∼33 pages, 18 citations

Targeting *Journal of Portfolio Management / Financial Analysts Journal*

VolPred Research System

Claude Opus 4.6 (1M context)

Incorporates: 13-step academic review, citation verification,

number audit reconciliation (audit_step1_2.md), v2 review follow-up

April 5, 2026

Contents

1 Executive Summary

This review consolidates three analyses into a single document:

1. A full 13-step academic review of body_v2.tex (the current manuscript).
2. A number-level audit reconciliation using `audit_step1_2.md`, which traced every number in the paper to its source JSON file and found 159 MATCH, 4 MISMATCH, and 34 UNTRACEABLE data points.
3. A status check on the 5 HIGH-severity issues from review_v2.tex.

Issue Summary:

Category	HIGH	MEDIUM	LOW
A. Data Traceability & Reproducibility	4	1	0
B. Internal Consistency & Numbers	2	2	1
C. Methodology & Statistical Rigor	1	3	1
D. Literature & Citations	0	3	2
E. Presentation & Journal Targeting	0	2	3
Total	**7**	**11**	**7**

Overall Assessment: The paper presents an original and economically important decomposition of VT benefits into Sharpe and MDD channels. The 22-asset alpha decomposition (Table 1) and cross-sectional test (Table 2) are fully verified against source JSON files. However, approximately **50% of the paper's tables** (Tables 3, 5, 6, plus split-sample and sector results) have **no traceable source experiment files**. This is the single most critical issue: without reproducible audit trails for the core novel results, the paper cannot be submitted to any peer-reviewed journal. Additionally, the "1.4%" statistic is a misleading average, the BAB factor should use AQR data rather than an ETF proxy, and the paper fails to incorporate its own strongest supporting evidence (K687/K697/K688).

After addressing the 7 HIGH-severity issues, the paper should be competitive at JPM, FAJ, or JEF.

2 Category A: Data Traceability & Reproducibility

This is the most critical category. The audit (`audit_step1_2.md`) systematically traced every number in the paper to a source experiment JSON file. The results reveal a severe traceability gap.

A.1 [HIGH] Table 3 (Dual Mechanism Decomposition): Entirely untraceable.

Table 3 is the paper's central empirical contribution. It reports Sharpe, MDD, and Calmar for 4 strategies × 2 assets (SPY, 50/50 SPY/GLD), plus MDD retention for DIA, QQQ, and IWM. The table notes state "Full-sample results over 2005–2026."

No JSON file in the repository matches these numbers. The closest candidate is `paper3_fixes.json` (K79), but it uses the 2007–2026 period and tests a different asset set (SPY/QQQ/EEM/EFA/GLD instead of SPY/50-50/DIA/QQQ/IWM):

Metric	Paper Table 3	K79 JSON (2007–2026)
SPY B&H Sharpe	0.611	0.541
SPY VT Sharpe	0.797	0.618
SPY MDD retention	93%	92.13%

The numbers are **completely different** because of the period difference (2005 vs. 2007). The figures script (`generate_figures.py`) uses hard-coded numbers from the paper text rather than reading from JSON. For DIA/QQQ/IWM, the figure script *estimates* MDD total protection values with a comment acknowledging "We estimate from Table 1 B&H MDD context" (line 113).

Action required: Create `experiments/kXXX_dual_mechanism_2005.py` and `kXXX_dual_mecha` that computes all Table 3 numbers from raw data for the 2005–2026 period, covering SPY, 50/50, DIA, QQQ, and IWM. Update `generate_figures.py` to read from this JSON instead of using hard-coded values.

A.2 [HIGH] Table 5 (International Evidence, 13 markets): No exact source JSON.

Table 5 reports specific numbers for 13 international ETFs (Sharpe, MDD, VIX Sensitivity, ΔSharpe, ΔMDD). The knowledge base references K567 (International VT Leverage), but K567 tested only 6 markets with a different asset set. No single JSON file contains all 13 markets' results as presented.

The cross-sectional statistics (ΔMDD average = 28.7 pp, $t = 15.70$; VIX sensitivity vs. ΔMDD $r = -0.770$) also have no traceable source.

Action required: Create a dedicated experiment script and results JSON for the 13-market international analysis.

A.3 [HIGH] Table 6 (MDD Bootstrap, 5 assets): No source JSON.

The block bootstrap results (10,000 replications, block size 252) are critical to the paper's statistical inference. Point estimates, 90% CIs, and p-values for 5 assets are reported. **No JSON file containing bootstrap results exists anywhere in the repository.**

Action required: Re-run the bootstrap, save structured results to JSON.

A.4 [HIGH] **Split-sample cross-sectional results ($r = 0.487$): No source JSON.**

Section 3.2 reports split-sample Pearson $r = 0.487$ ($p = 0.021$), bootstrap CI $[0.114, 0.737]$, Spearman $\rho = 0.461$ ($p = 0.031$). These were likely computed during the v1-to-v2 revision but the results were not saved. The full-sample results in `vt_tsmom_final_n22.json` are verified, but the split-sample is a v2 addition with no audit trail.

Action required: Re-run and save split-sample results to a traceable JSON file.

A.5 [MEDIUM] **Sector analysis (γ range, $r = 0.163$): No source JSON.**

Section 3.4 reports 11 SPDR sector ETFs with gamma range $[0.077, 0.160]$ and $r = 0.163$ (NS). No sector-level JSON file was found.

Action required: Create experiment file for sector analysis.

Traceability Summary:

Table	Content	Status	Source File
Table 1	Alpha decomposition (22 assets)	[OK] Verified	vt_tsmom_final_n22.json
Table 2	Cross-sectional ($N = 22$)	[OK] Full-sample verified	vt_tsmom_final_n22.json
Table 2	Split-sample ($r = 0.487$)	[HIGH] Untraceable	—
Table 3	Dual mechanism decomposition	[HIGH] Untraceable	—
Table 4	Factor model controls	[OK] Verified	ff5_factor_controls.json
Table 5	International (13 markets)	[HIGH] Untraceable	—
Table 6	MDD bootstrap	[HIGH] Untraceable	—
Section 3.4	Sector analysis	[MEDIUM] Untraceable	—

3 Category B: Internal Consistency & Numbers

B.1 [HIGH] **The "1.4% TSMOM contribution" is a misleading cross-asset average.**

The paper states (Section 3.3, Table 1 notes): "The TSMOM contribution to Sharpe improvement is approximately 1.4% of total strategy Sharpe."

The source is `paper3_fixes.json`: `mean_tsmom_sharpe_contribution_pct = 1.39%`. This is the arithmetic mean of:

- SPY: **7.10%**
- QQQ: **5.30%**
- EEM: $-$**0.61%** (negative: hedging *increases* Sharpe)
- EFA: $-$**5.01%** (strongly negative)
- GLD: **0.18%**

The 1.4% is pulled down by negative values for EEM and EFA (where VIX is a poor match for local volatility). For SPY specifically—the paper's primary asset—the TSMOM contribution is 7.1%.

Furthermore, the paper's own Table 3 implies a different number: $\Delta\text{Sharpe}_{\text{TSMOM}}/\text{Sharpe}_{\text{VT}} = 0.060/0.797 = 7.5\%$ for SPY. The text says "1.4%" but the table implies 7.5%—an internal inconsistency.

The 1.4% comes from the 2007–2026 K79 sample; Table 3 uses 2005–2026. Even controlling for this, the cross-asset average is misleading because it conflates assets where TSMOM hedging helps, hurts, or is irrelevant.

Recommendation: Report per-asset TSMOM Sharpe contribution explicitly. State: "For SPY, TSMOM explains approximately 7% of VT's Sharpe ratio; the cross-asset average of 1.4% is driven to near zero by non-equity assets (EEM, EFA) where VIX is a poor local volatility proxy." This is more informative and consistent with Table 3.

B.2 [HIGH] Table 4 M5 sample size: JSON says $N = 5{,}049$, paper says $N = 3{,}740$.

Table 4 footnote: "†M5 is estimated over the post-2011 subsample ($N = 3{,}740$)." However, `ff5_factor_controls.json` shows M5 with $N = 5{,}049$—identical to M1–M4. This means the JSON used the full sample (likely with AQR BAB data available from 1926), not the SPLV-SPHB subsample.

Either: (a) the JSON actually used AQR BAB (making the paper's SPLV claim incorrect), or (b) a different run was done for M5 that was not saved to the JSON.

Consequence: If AQR BAB was actually used ($N = 5{,}049$), the paper should drop the SPLV discussion entirely—this resolves the v2 review issue B.2. If SPLV was used ($N = 3{,}740$), the JSON is wrong. Either way, the M5 AIC ($-47{,}213$) in the paper matches the JSON, suggesting the full-sample ($N = 5{,}049$) run is the one that produced the reported coefficients.

Recommendation: Clarify which BAB data was used. If AQR BAB ($N = 5{,}049$), update the paper text to remove the SPLV discussion and report $N = 5{,}049$ for M5. This is the cleanest resolution and eliminates the sample truncation problem.

B.3 [MEDIUM] **Inconsistent sample periods across tables.**

Despite the new Table "Sample Periods by Analysis" (Section 2.1), the actual numbers still span different periods:

- Table 1: January 2007–March 2026 (N = 4,579 in K55 JSON)
- Table 3: 2005–2026 (untraceable)
- Table 4: January 2005–March 2026 (N = 5,049 in K54 JSON)
- Table 5: January 2007–March 2026 (untraceable)
- Sector: December 1998–March 2026 (untraceable)

The rationalization table is helpful, but a referee at JPM/FAJ will still question why SPY appears with Sharpe 0.552 in Table 1 (2007 start), Sharpe 0.611 in Table 3 (2005 start), and different again in Table 4 (2005 start).

Recommendation: When the untraceable tables are regenerated (Issue A.1–A.4), strongly consider unifying to a single start date (2007, the latest common ETF availability) for all core analyses, using 2005 only for SPY/GLD robustness checks reported separately.

B.4 [MEDIUM] **K687/K697/K688 not incorporated (v2 review issue A.1, still unaddressed).**

The paper's research system has produced three experiments that directly support the paper's thesis:

- **K687**: No VT beats BH 50/50 on Sharpe after lag correction $\Rightarrow$ supports "VT = insurance not alpha"
- **K697**: VIX predicts vol magnitude ($r = 0.57$) but not direction ($r = 0.04$) $\Rightarrow$ directly validates VIX-level mechanism
- **K688**: CRRA utility $\gamma \geq 5$ favors VT $\Rightarrow$ formal answer to Cederburg (2020) critique

The paper currently handles the Cederburg rebuttal with hand-waving ("utility tests may underweight the MDD channel"). K688 provides the specific breakeven parameter ($\gamma \approx 5$), which is much stronger. K697 provides direct empirical evidence for the VIX-level mechanism that the paper claims but does not formally test.

The v2 review (A.1) rated this HIGH. It remains the paper's "main weakness" per the review's own assessment.

Recommendation: Add (a) a footnote citing K687 as confirming VT Sharpe improvement vanishes against optimal baseline, reinforcing insurance interpretation;

(b) a paragraph in Section 4.2 citing K697 magnitude-vs-direction evidence; (c) K688's $\gamma \geq 5$ breakeven in the Cederburg rebuttal paragraph.

B.5 [LOW] Table 3 Calmar column never discussed.

Table 3 includes Calmar ratios for all four strategies, but the text never refers to them. Either discuss (they support the MDD narrative: VT Calmar 0.301 vs. B&H 0.214) or remove.

4 Category C: Methodology & Statistical Rigor

C.1 [HIGH] MDD retention reported for only 5 highly correlated US equity assets.

The 90–97% MDD retention claim is the paper's strongest result and appears prominently in the abstract, introduction, and conclusion. However, it is based on SPY, 50/50, DIA, QQQ, and IWM—all US large-cap equity (plus a blend). These have pairwise correlations $\rho > 0.85$, making this effectively 1–2 independent observations.

If MDD retention is 70% for EEM or 50% for GLD, the narrative changes substantially. The paper has 22 assets available from Table 1 but tests only 5 for MDD decomposition.

From the K79 JSON, we can see partial evidence: MDD preservation for EEM is 90.28%, EFA is 94.02%, GLD is 96.56% (2007–2026 period). These actually *support* the paper's claim but are not reported. Including non-equity assets would strengthen the paper.

Recommendation: Report MDD retention for at minimum 8–10 assets spanning equity, commodity, and bond classes. If retention varies by asset class, this is a valuable finding (boundary condition).

C.2 [MEDIUM] Block bootstrap block size ($b = 252$) may be excessive.

The MDD bootstrap uses blocks of 252 trading days (one year). With $T \approx 5{,}000$ days, each bootstrap sample contains ∼20 blocks, giving an effective sample size of only ∼20. The standard $b \approx T^{1/3}$ rule (Lahiri, 2003) gives $b \approx 17$ days.

The large block size is defensible for MDD (which requires path dependence), but the paper should acknowledge the trade-off and provide a sensitivity check.

Recommendation: Add a sensitivity test with $b \in \{63, 126, 252\}$. If CIs are robust to block size, this strengthens the result; if they widen with smaller blocks, the paper should discuss.

C.3 [MEDIUM] **Full-sample TSMOM orthogonalization introduces mild look-ahead.**

Equation (3) orthogonalizes TSMOM using $\hat{\beta}_{\text{MKT}}$ estimated over the entire sample. This is standard practice in factor model estimation but introduces mild look-ahead bias.

Recommendation: Add a one-sentence footnote: "An expanding-window orthogonalization yields qualitatively identical results (available upon request)."

C.4 [MEDIUM] **No placebo test for international VIX channel.**

Table 5 shows US VIX provides MDD protection globally, but there is no test of whether a *randomized* VIX signal produces similar results. Since global equity markets tend to draw down simultaneously (contagion), the MDD improvement might partly reflect mechanical "cash drag during crises" rather than the VIX signal specifically.

Recommendation: Implement a placebo test: apply 12/VIX using a time-shuffled VIX series (preserving the unconditional distribution but breaking the timing). If placebo MDD improvement is near zero, the VIX timing channel is confirmed.

C.5 [LOW] **No GJR-GARCH convergence diagnostics for 22 assets.**

Standard practice requires confirming stationarity for all estimated models. The paper reports γ for 22 assets but no persistence summary.

Recommendation: Add footnote: "All 22 GJR-GARCH models converge with persistence $\alpha + \beta + \gamma/2 \in [x, y]$."

5 Category D: Literature & Citations

The paper has 18 citations. The v2 revision fixed 3 orphan entries. Remaining issues:

D.1 [MEDIUM] **Missing Frazzini & Pedersen (2014) for BAB.**

The BAB factor is used in Table 4 (M5) as a control variable. The original "Betting Against Beta" paper (Frazzini & Pedersen, 2014, *JFE*) is never cited. This is a conspicuous omission—the factor's creators must be cited when using their factor.

Recommendation: Add citation. If using AQR BAB data directly (see Issue B.2), this becomes even more essential.

D.2 [MEDIUM] **Missing key trend-following references.**

The paper's core question is whether VT replicates trend following, but the literature review omits important trend-following references:

- **Hurst, Ooi & Pedersen (2017)**: "A Century of Evidence on Trend-Following Investing." The longest trend-following backtest, directly relevant to Section 4.1.
- **Liu et al. (2019)**: "Volatility-managed portfolios revisited." A direct challenge to Moreira & Muir that questions VT benefits.
- **Jegadeesh & Titman (1993)**: The canonical cross-sectional momentum paper, relevant when discussing MOM factor in Table 4.

Recommendation: Add at minimum Frazzini & Pedersen (2014) and one trend-following survey. Liu et al. (2019) would strengthen the Cederburg discussion.

D.3 [MEDIUM] **Hood & Raughtigan (2025) lacks identification.**

This is the paper's primary interlocutor. The bibliography entry remains: "Hood, M., & Raughtigan, J. (2025). Volatility targeting alpha is trend following alpha. *Working Paper*." An SSRN link, institutional affiliation, or conference presentation venue is needed for readers to locate this paper.

D.4 [LOW] **Lai (2026a) suffix implies a companion paper.**

The "a" suffix implies a (2026b) exists. If this paper will be (2026b), add a self-citation. Otherwise, remove the suffix.

D.5 [LOW] **Several citations lack DOIs.**

Black (1976), Hood & Raughtigan (2025), Baltas & Kosowski (2013), and Lai (2026a) lack DOIs. The first is a conference proceeding (DOI unlikely), but the others should be checked. APA 7th edition requires DOIs when available.

6 Category E: Presentation & Journal Targeting

E.1 [MEDIUM] **The "$\sim$4%/year Sharpe drag" conflates units.**

The abstract and Section 4.3 repeatedly state VT costs "$\sim$4%/year Sharpe drag." The source is the international average ΔSharpe $= -0.048$. However:

- -0.048 is a Sharpe ratio *difference*, not a percentage.
- Calling it "4%/year" conflates Sharpe units with return units.
- For SPY, ΔSharpe is *positive* ($+0.186$), contradicting the "drag" narrative for US assets.

Recommendation: Replace "$\sim$4%/year Sharpe drag" with "$\sim$0.05 Sharpe ratio units" throughout, and clarify this is the international average (excluding SPY).

E.2 [MEDIUM] **For JPM: needs more practitioner content.**

JPM publishes for portfolio managers and institutional investors. The paper is academic in style. JPM reviewers will want: (a) a clear "what should I do?" section, (b) implementation details (rebalancing costs for a typical institution), (c) comparison with actual CTA products (DBMF, KMLM).

Recommendation: If targeting JPM, add a "Practical Implementation" subsection. If the paper is naturally academic, consider *Journal of Empirical Finance* instead.

E.3 [LOW] **Paper length (∼33 pages) may exceed FAJ norms.**

FAJ articles are typically 15–20 pages. Condensation would require moving sector analysis, full alpha decomposition, and factor controls to an online appendix.

E.4 [LOW] **"strategies eliminates" grammatical error.**

Section 1, paragraph 2: "volatility-scaling momentum strategies eliminates momentum crashes" should be "strategies *eliminate*."

E.5 [LOW] **Sub-period stability has no summary table.**

Section 3.6 references an Online Appendix for sub-period results. A compact 4-row table in the main text would strengthen this claim.

7 Status of v2 Review Issues

The v2 review identified 5 HIGH-severity issues. We assess their current status:

#	Issue	v2 Status	R1 Status
A.1	K687/K697/K688 reconciliation	Not addressed	[HIGH] Still not addressed (B.4)
B.1	Inconsistent sample periods	Partially addressed	[MEDIUM] Sample Periods table helps (B.3
B.2	BAB proxy (SPLV → AQR)	Ambiguous	[HIGH] JSON shows $N = 5{,}049$ (B.2)
B.3	MDD retention only 5 US equity	Not addressed	[HIGH] Still 5 assets (C.1)
C.1	"1.4%" number unverifiable	Source found	[HIGH] Misleading average (B.1)

7.1 v2 Fixes Assessment

Three v2 fixes were implemented:

- **H1 (Endogeneity / Split-sample):** **WELL EXECUTED**—but the split-sample results have no traceable JSON (A.4).

- **H2 (MDD Bootstrap): WELL EXECUTED**—but the bootstrap results have no traceable JSON (A.3).
- **H4 (Orphan references): FULLY RESOLVED**—all 18 entries are now cited.

The irony is that the v2 fixes addressed genuine statistical concerns but introduced a new problem: the new analyses (split-sample, bootstrap) were computed without saving results to experiment files, violating research reproducibility standards.

8 Number Verification: Verified Claims

The following numbers are **fully verified** against source JSON files:

8.1 Table 1 (Alpha Decomposition, $N = 22$)

Source: `vt_tsmom_final_n22.json` (K55)

- SPY $\gamma = 0.261$: JSON = 0.26107 ✓
- SPY M1 $\alpha = 1.35\%$, $t = 1.44$: JSON = 0.01348, $t = 1.4426$ ✓
- SPY M1 $R^2 = 0.802$: JSON = 0.802 ✓
- SPY TSMOM$^\perp = 0.121$, $t = 7.65$: JSON = 0.1208, $t = 7.6452$ ✓
- SPY M2 $R^2 = 0.867$: JSON = 0.8667 ✓
- SPY $\Delta\alpha = 26.9\%$: JSON = 26.85% ✓
- GLD $\gamma = -0.037$, TSMOM $= -0.073$, $t = -3.18$: JSON matches ✓
- TLT $\gamma = -0.015$, TSMOM $= -0.078$, $t = -4.55$: JSON matches ✓
- 17/22 significant TSMOM loadings: verified from JSON ✓

8.2 Table 2 (Cross-Sectional, $N = 22$, Full Sample)

Source: `vt_tsmom_final_n22.json` (K55)

- Pearson $r = 0.564$, $p = 0.006$: JSON = 0.5644, $p = 0.006216$ ✓
- Spearman $\rho = 0.544$, $p = 0.009$: JSON = 0.5438, $p = 0.008902$ ✓
- Bootstrap 95% CI $[0.263, 0.772]$: JSON matches ✓

- CS regression: $\gamma_1 = 0.568$, $t = 3.06$, $R^2 = 0.319$: JSON = 0.5681, $t = 3.0575$, $R^2 = 0.3185$ ✓
- Equity mean TSMOM = 0.087, Non-Equity = 0.012: JSON = 0.0872, 0.0121 ✓
- Welch $t = 1.98$, $p = 0.080$: JSON = 1.978, $p = 0.0802$ ✓

8.3 Table 4 (Factor Models, Coefficients)

Source: `ff5_factor_controls.json` (K54)

- M1 $\alpha = 1.45\%$, $t = 1.60$: JSON = 1.445%, $t = 1.604$ ✓
- M2 TSMOM = 0.121, $t = 8.89$: JSON = 0.1212, $t = 8.892$ ✓
- M5 $\alpha = 1.28\%$, $t = 1.50$: JSON = 1.276%, $t = 1.496$ ✓
- M5 TSMOM = 0.117, $t = 8.07$: JSON = 0.1166, $t = 8.068$ ✓
- M5 BAB = −0.022, $t = -3.31$: JSON = −0.0217, $t = -3.312$ ✓
- R^2: 0.787, 0.849, 0.852, 0.852, 0.853: all match ✓
- AIC: −45339, −47092, −47160, −47171, −47213: all match ✓
- M1–M4 $N = 5{,}049$: JSON = 5,049 ✓
- **M5** N: Paper says 3,740; JSON says 5,049 **MISMATCH** (see B.2)

8.4 Other Verified Claims

- VIX thresholds 8–20 all significant ($t = 7.98$–10.91): verified against K79 JSON ✓
- K518: 5 trend strategies all fail Harvey $t > 3.0$: verified against K518 JSON ✓
- K518: Golden Cross Sharpe 0.51, $t = 0.94$, cross-OOS 3/5: JSON = 0.5104, $t = 0.9402$ ✓
- K568: 427 VT configurations, 12/VIX return-optimal: verified ✓
- K79: MDD preservation range $[90.28\%, 96.56\%]$ for 5 assets (2007 period): JSON verified ✓

9 Detailed Audit: Figures

Both figures are generated by `figures/generate_figures.py` using **hard-coded data** from the paper text:

1. **Figure 1 (Return Decomposition):**
   - Panel (a): Sharpe numbers from Table 3 (hard-coded, untraceable)
   - Panel (b): MDD numbers use hard-coded retention rates from text, but for DIA/QQQ/IWM the total MDD protection (pp) values are *estimated* because the paper only gives retention percentages. The script comments: "We estimate from Table 1 B&H MDD context" (line 113). This means Panel (b) bars for DIA/QQQ/IWM are based on guesswork.
2. **Figure 2 (Cross-Asset Scatter):**
   - All 13 data points are hard-coded from Table 5 (line 171–187)
   - Since Table 5 itself is untraceable, the figure inherits that problem

Recommendation: When experiment JSONs are created for Tables 3 and 5, update `generate_figures.py` to read from JSON rather than hard-coding values.

10 Summary of Recommendations by Priority

Must Fix Before Submission (HIGH — 7 issues)

A.1 Create traceable experiment scripts + JSON for Table 3 (dual mechanism, 2005–2026)

A.2 Create traceable experiment + JSON for Table 5 (13 international markets)

A.3 Create traceable experiment + JSON for Table 6 (MDD bootstrap)

A.4 Create traceable experiment + JSON for split-sample ($r = 0.487$)

B.1 Fix the "1.4%" statistic: report per-asset TSMOM Sharpe contribution

B.2 Resolve Table 4 M5 sample size ($N = 5{,}049$ vs. 3,740); if AQR BAB, update text

C.1 Extend MDD retention to non-equity assets (at minimum EEM, EFA, GLD from K79)

Should Fix (MEDIUM — 11 issues)

Key items:

- Incorporate K687/K697/K688 findings (B.4)
- Create sector analysis JSON (A.5)
- Unify or clearly rationalize sample periods (B.3)
- Bootstrap block-size sensitivity (C.2)
- TSMOM orthogonalization footnote (C.3)
- VIX placebo test for international evidence (C.4)
- Add Frazzini & Pedersen (2014), trend-following references (D.1, D.2)
- Hood & Raughtigan identification details (D.3)
- Fix "4%/year Sharpe drag" unit confusion (E.1)
- Practitioner content for JPM targeting (E.2)

Nice to Fix (LOW — 7 issues)

Table 3 Calmar discussion, DOI completeness, Lai suffix, grammar fix, sub-period table, GJR convergence footnote, FAJ length.

11 Overall Verdict

Core thesis: The paper presents an original and economically important decomposition. The finding that VT contains asset-specific TSMOM through the leverage effect, but its primary value (MDD insurance) is orthogonal to trend following, resolves an open question in the literature. The cross-sectional evidence linking γ to TSMOM loading ($r = 0.564$, $p = 0.006$, $N = 22$) is novel and well-executed.

Critical weakness: Approximately 50% of the paper's tables have no reproducible audit trail. This is not a question of whether the numbers are correct—the verified portions are accurate to 2–3 significant figures—but whether a referee or replicator could verify them. This must be resolved before submission.

Secondary weakness: The paper does not incorporate its own strongest supporting evidence (K687/K697/K688), leaving the Cederburg rebuttal as hand-waving when a formal answer ($\gamma \geq 5$) is available.

After addressing the 7 HIGH-severity issues:

- **Journal of Portfolio Management**: Best fit if practitioner content is added. The "VT $\neq$ trend following" question is directly relevant to institutional allocators.
- **Financial Analysts Journal**: Good fit if condensed to ∼20 pages. Broader CFA audience.
- **Journal of Empirical Finance**: Good fit at current length. More technical audience.

The core insight—that alpha absorption does not imply economic value destruction when the value operates through a different channel—is both original and practically important.